Solution Requirements

Team ID: UNM-293 2026-2430

Project Name: FitBuddy – AI Fitness Plan Generator using Gemini Models

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Maximum Marks: 4 Marks

The following requirements are prepared using the structure of the provided Solution Requirements document. Functional Requirements describe what FitBuddy should do, while Non-Functional Requirements describe how well it should operate.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users should be able to securely sign in or verify their identity before accessing personalized fitness-plan features.	High
2	Authorization levels	The system should provide appropriate access permissions for users and administrators. Users access their plans and profiles; authorized administrators manage system content and controls.	Medium
3	External interfaces	FitBuddy should connect to Gemini Models through an API to generate personalized fitness-plan recommendations. It may also use a database for storing user profiles and plans.	High
4	Transactions processing	The system should process user inputs such as fitness goals, preferences, activity level and available time, then send the required information for plan generation and return the generated plan.	High
5	Reporting	The system should display generated fitness plans and user progress information in a clear dashboard or report format. Plans should be available for later viewing.	Medium
6	Business rules, etc.	Fitness plans should be generated according to the user's stated goals, preferences, activity level and available time. Recommendations should be personalized and practical.	High
7	Compliance to laws or regulations	The system should protect personal information, use appropriate privacy practices, and avoid presenting AI-generated fitness information as a substitute for professional medical advice.	High
8	Other	Users should be able to enter or update fitness information and regenerate or modify plans when their requirements change.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should respond quickly when users submit fitness information and request a generated plan.	Normal application pages should load within about 2 seconds; plan-generation requests should provide a response or clear status within an acceptable waiting period.
2	Scalability	The architecture should allow the application to support increasing numbers of users and stored fitness plans.	System should support growth in users and data without major redesign; API usage should be managed efficiently.
3	Security & Data Privacy	User fitness and profile information should be protected from unauthorized access. Data should be transmitted and stored securely.	Use authenticated access, HTTPS/TLS for data transfer, secure storage practices and appropriate access controls.
4	Reliability & Availability	FitBuddy should remain available and handle temporary API or service failures gracefully.	Provide clear error messages and recovery/retry handling where appropriate; target high availability during normal operation.
5	Usability & Accessibility	The interface should be simple, understandable and easy to navigate for users with different levels of technical knowledge.	Users should be able to enter details, generate a plan and view results with minimal steps; use readable text, clear controls and accessible design.
6	Maintainability & Portability	The software should use a modular structure so that Gemini integration, user management and fitness-plan features can be updated independently.	Code should be organized into maintainable modules and documented sufficiently for future updates or deployment on supported environments.